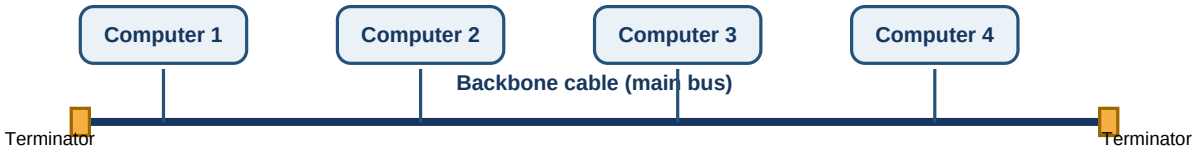
Network Topologies: Answers

1. Draw a bus topology diagram, label its key components, and give one real-world example.

Answer: A bus topology connects multiple computers to one shared main cable called the **backbone**. Each computer joins the backbone through a short connection (drop line). **Terminators** are placed at both ends of the backbone to absorb signals and help prevent signal reflections.



Short drop lines connect each device to the shared cable.

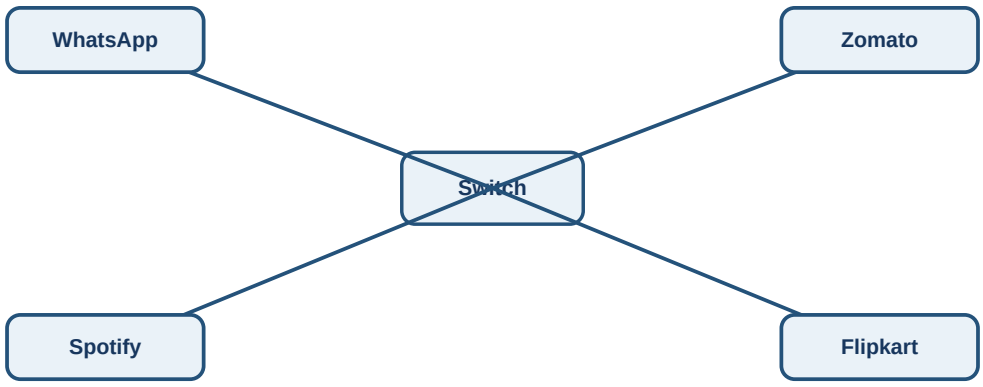
Key components

- **Computers/devices:** send and receive data on the network.
- **Backbone cable:** the shared transmission path used by all connected devices.
- **Drop lines/taps:** connect each device to the backbone.
- **Terminators:** fitted at both cable ends to reduce signal reflections.

Real-world example: Older small Ethernet LANs using 10BASE2 (thin coaxial cable) or 10BASE5 (thick coaxial cable) used bus-style cabling. This is mainly a historical example because modern office networks usually use switches and star-shaped cabling.

2. Create a star topology in Cisco Packet Tracer with one switch and four PCs named after popular apps, then take a screenshot.

Answer and step-by-step procedure: The diagram below shows the intended Packet Tracer layout. Recreate it in Cisco Packet Tracer and take a screenshot of your own workspace; a real screenshot must be captured from the application, so this PDF includes a clearly labelled reference diagram rather than pretending it is an actual Packet Tracer screenshot.



Star topology: each PC has its own link to the central switch.

Steps in Cisco Packet Tracer

Step 1. Open Cisco Packet Tracer and create a new, blank workspace.

Step 2. From **Network Devices** → **Switches**, drag one switch (for example, **2960**) into the workspace.

Step 3. From **End Devices**, drag four PCs into the workspace.

Step 4. Rename the PCs to **WhatsApp**, **Zomato**, **Spotify**, and **Flipkart**. Depending on the Packet Tracer version, click the device name/label or open the device and edit its display name.

Step 5. Choose **Connections** (lightning-bolt icon), then select **Copper Straight-Through**. Connect each PC's **FastEthernet0** port to a separate switch port, such as **Fa0/1** through **Fa0/4**.

Step 6. Wait for the link indicators to turn green. If they remain amber briefly, allow spanning/initialization time; check the selected ports and cable if a link stays down.

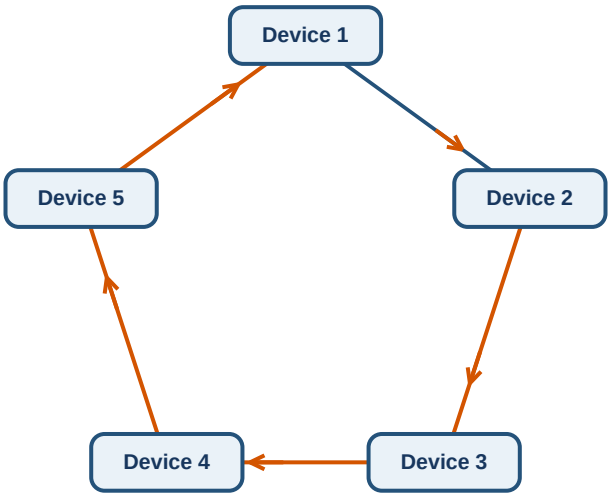
Step 7. Optional connectivity test: open each PC's **Desktop** → **IP Configuration** and assign addresses in one subnet, for example 192.168.1.1, .2, .3, and .4 with subnet mask 255.255.255.0. Then use **Desktop** → **Command Prompt** on one PC and ping another PC's address.

Step 8. Arrange the four PCs around the switch so the star shape is clear. Save the project using **File** → **Save As**.

Step 9. Capture the requested screenshot: use your operating system's screenshot shortcut (for example, **Win + Shift + S** on Windows), select the Packet Tracer workspace, and save or paste the screenshot into your assignment if required.

3. Draw a ring topology and explain how data travels.

Answer: In a ring topology, each device is connected to two neighboring devices, forming a closed loop. Data moves from one device to the next around the ring in the direction shown by the arrows; each device forwards the data until it reaches the intended destination. Some ring networks use a token or other access-control method to coordinate which device may transmit.



Arrows show one possible direction of data flow.

How to read the diagram

The arrows indicate one possible clockwise direction of travel. For example, data sent by Device 1 can pass through Device 2, then Device 3, and continue around the loop until it reaches its destination. The exact direction and forwarding method depend on the ring technology; some implementations can support traffic in both directions or provide a recovery path.

4. Compare mesh and star topologies: two advantages and two disadvantages of each.

A **mesh topology** provides multiple direct links between devices (a full mesh connects every device to every other device). A **star topology** connects each end device to a central device, usually a switch. The table compares typical characteristics.

Topology	Advantages (2)	Disadvantages (2)
Mesh	1. **High reliability:** multiple paths can keep communication working if one link fails. 2. **Good fault isolation:** a failed link is often easier to identify without taking down every connection.	1. **Expensive:** many cables and ports are required; a full mesh grows rapidly as devices are added. 2. **Complex to install and maintain:** the large number of links makes cabling and configuration more difficult.
Star	1. **Easy to install and expand:** new devices can be connected to an available switch port. 2. **Easy troubleshooting:** a single cable or port failure usually affects only the connected device.	1. **Central-device dependency:** if the switch/hub fails, connected devices may lose network communication. 2. **More cabling than a bus:** each device needs its own cable run to the central switch.

Conclusion: Mesh is often selected when resilience is especially important and the cost/complexity is acceptable. Star is common in modern LANs because switches make it straightforward to connect, manage, and expand devices; its main weakness is reliance on the central switch.

Quick revision

• **Bus:** one shared backbone cable with terminators at both ends.
• **Star:** every end device has a separate link to a central switch/hub.
• **Ring:** devices form a loop and forward data around it.
• **Mesh:** multiple interconnections provide alternate paths but increase cost and complexity.